

Resource Constraints Knowledge Base

Enabling Autonomous Agentic Action in Emotional Infrastructure

Executive Summary

The **Structured Experience Unit (SEU)** provides the perception and decision substrate for agentic AI—enabling agents to see, understand, and prioritize emotional truth structurally. However, **perception and prioritization alone cannot drive autonomous action.**

To move from "what matters most" to "what should we do given real-world limits," agents require a **Resource Constraints Knowledge Base**—a structured repository of organizational capacity, dependencies, risks, and strategic priorities that transforms the SEU from a decision substrate into a **portfolio optimization engine.**

This document defines the architecture, content, and rationale for the Resource Constraints Knowledge Base, making fully autonomous agentic action possible.

The Problem: Why Resource Constraints Matter

Without Resource Constraints

An agent powered by SEUs can:

- ☒ Perceive emotional truth (unpack raw feedback into structured intelligence)
- ☒ Decide what matters (rank drivers by Priority + Temporal Intelligence)
- ☒ Recommend action (generate OUs, problem statements, PDCA structures)

But it cannot answer:

- "We have 12 PO drivers. Budget for 3. Which 3?"

- "Should we fix Triage Efficiency (\$150K, elasticity 2.1) or Call Bell System (\$80K, elasticity 1.8)?"
- "Can we deploy both interventions in Q1, or will staff capacity limit us?"
- "If we fix X, will it worsen Y? Should we bundle them?"
- "This driver has regulatory risk. Does that override ROI logic?"

The gap: SEUs provide **priority** (consequence tiers: P0-P5). Resource constraints provide **feasibility** (what's actually achievable given limits).

With Resource Constraints

The agent becomes a **strategic portfolio optimizer** that:

- Selects the optimal set of interventions within budget and capacity limits
- Bundles interventions when dependencies exist
- Sequences actions across time horizons (Q1, Q2, Q3, Q4)
- Weights risk (regulatory, reputational, churn) against ROI
- Adapts mid-cycle when plans fail or new crises emerge

This transforms the agent from tactical (act on this driver) to strategic (optimize the entire portfolio).

The Architecture: What the Knowledge Base Contains

The Resource Constraints Knowledge Base consists of **seven interconnected modules**, each providing critical decision intelligence for autonomous action.

Module 1: Financial Constraints

Purpose

Define budgetary limits and cost structures that govern intervention selection and portfolio optimization.

Required Data

1.1 Budget Allocation

budget_allocation:

fiscal_year: 2025

total_annual_budget: 2000000

by_period:

Q1:

amount: 500000

start_date: "2025-01-01"

end_date: "2025-03-31"

Q2:

amount: 300000

start_date: "2025-04-01"

end_date: "2025-06-30"

Q3:

amount: 600000

start_date: "2025-07-01"

end_date: "2025-09-30"

Q4:

amount: 600000

start_date: "2025-10-01"

end_date: "2025-12-31"

by_category:

fix_interventions: 800000

optimize_interventions: 600000

innovate_initiatives: 400000

amplify_programs: 200000

reserve_buffer: 0.10 # 10% emergency reserve

1.2 Cost Models (By Intervention Type)

cost_models:

fix:

patient_safety:

min: 50000

max: 200000

avg: 120000

lead_time_days: 90

clinical_services:

min: 80000

max: 250000

avg: 150000

lead_time_days: 120

support_quality:

min: 30000

max: 100000

avg: 60000

lead_time_days: 60

optimize:

process_efficiency:

min: 40000

max: 120000

avg: 75000

lead_time_days: 90

communication:

min: 20000

max: 80000

avg: 45000

lead_time_days: 45

innovate:

technology:

min: 100000

max: 500000

avg: 250000

lead_time_days: 180

service_design:

min: 60000

max: 200000

avg: 120000

lead_time_days: 120

amplify:

advocacy_programs:

min: 15000

max: 60000

avg: 30000

lead_time_days: 30

1.3 Variable Cost Factors

cost_modifiers:

by_es_band:

ES5_Critical_Failure: 1.4 # 40% higher due to urgency

ES4_Active_Crisis: 1.2

ES3_At_Risk: 1.0

ES2_Growth_Opportunity: 0.9

ES1_Secure_Loyalty: 0.8

by_scope:

enterprise_wide: 1.5

multi_department: 1.2
single_department: 1.0
single_unit: 0.8

by_complexity:
high: 1.3
medium: 1.0
low: 0.7

Why This Matters

- **Portfolio optimization:** Agent can select interventions that maximize elasticity per dollar spent
 - **Period planning:** Agent knows what's affordable in Q1 vs Q2
 - **Trade-off reasoning:** Agent can compare "\$150K for elasticity 2.1" vs "\$80K for elasticity 1.8"
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Module 2: Capacity Constraints

Purpose

Define human capital, operational capacity, and deployment bandwidth that limit simultaneous intervention execution.

Required Data

2.1 Staff Capacity

staff_capacity:
total_fte: 25

by_role:
project_managers:
fte: 5
hours_per_week: 40
max_concurrent_projects: 2

clinical_leads:
fte: 8
hours_per_week: 40
max_concurrent_projects: 1

process_improvement:

fte: 6
hours_per_week: 40
max_concurrent_projects: 3

change_management:
fte: 4
hours_per_week: 40
max_concurrent_projects: 2

data_analysts:
fte: 2
hours_per_week: 40
max_concurrent_projects: 4

2.2 Deployment Bandwidth

deployment_constraints:
max_concurrent_interventions: 8
max_per_department:
emergency_services: 2
inpatient_care: 3
ambulatory_services: 2
support_services: 2

blackout_periods:
- name: "Joint Commission Review"
start: "2025-05-01"
end: "2025-05-15"
affected_themes: ["Patient Safety", "Clinical Services"]

- name: "Budget Freeze"
start: "2025-12-15"
end: "2025-12-31"
affected_streams: ["all"]

2.3 Intervention Resource Requirements

resource_requirements:
by_intervention_type:
fix_patient_safety:
project_manager_hours: 120
clinical_lead_hours: 200
process_improvement_hours: 80
change_management_hours: 60
duration_weeks: 12

fix_clinical_services:
project_manager_hours: 160
clinical_lead_hours: 240
process_improvement_hours: 120
duration_weeks: 16

optimize_communication:
project_manager_hours: 80
process_improvement_hours: 100
change_management_hours: 120
duration_weeks: 8

Why This Matters

- **Feasibility checking:** Agent knows if it can actually deploy 8 interventions or if staff capacity limits it to 5
 - **Temporal sequencing:** Agent can stagger interventions to avoid overloading teams
 - **Blackout awareness:** Agent won't schedule patient safety fixes during Joint Commission reviews
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Module 3: Risk Models & Regulatory Constraints

Purpose

Define risk thresholds, regulatory exposure, and compliance timelines that may override pure ROI optimization.

Required Data

3.1 Regulatory Risk

regulatory_constraints:
patient_safety:
governing_body: "Joint Commission"

triggers:
ES5_Critical_Failure:
action_required_within_days: 90
penalty_if_unaddressed: "Accreditation risk"

ES4_Active_Crisis:

action_required_within_days: 180
penalty_if_unaddressed: "Citation"

priority_override: true # Regulatory risk overrides ROI

hipaa_compliance:
governing_body: "HHS Office for Civil Rights"

triggers:
data_breach_risk:
action_required_within_days: 30
penalty_range: [100000, 1500000]

priority_override: true

3.2 Reputational Risk

reputational_risk:
public_facing_drivers:
- "Emergency Services → Triage Efficiency"
- "Support Quality → Communication Style"
- "Billing Experience → Transparency"

risk_multipliers:
ES5_Critical_Failure: 3.0 # 3× media exposure risk
ES4_Active_Crisis: 2.0
ES3_At_Risk: 1.5

social_media_amplification:
emergency_services: 2.5
patient_safety: 3.0
billing_disputes: 2.0

3.3 Churn Risk Models

churn_risk:
by_es_band_and_trajectory:
ES5_stable:
churn_probability_6mo: 0.75
revenue_at_risk_per_patient: 15000

ES4_declining:
churn_probability_6mo: 0.60
revenue_at_risk_per_patient: 12000

ES3_declining:

churn_probability_6mo: 0.40

revenue_at_risk_per_patient: 8000

ES3_stable:

churn_probability_6mo: 0.20

revenue_at_risk_per_patient: 5000

high_value_patient_threshold: 50000 # Annual revenue

Why This Matters

- **Risk-weighted prioritization:** ES5 + Patient Safety gets prioritized even if another driver has higher elasticity
 - **Compliance urgency:** Agent knows "must address within 90 days" and plans accordingly
 - **Revenue protection:** Agent factors churn probability × revenue-at-risk into portfolio decisions
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Module 4: Dependency Models & System Dynamics

Purpose

Define cross-driver dependencies, feedback loops, and bundling opportunities that affect intervention design.

Required Data

4.1 Cross-Driver Dependencies

dependencies:

causal_relationships:

- driver_a: "Emergency Services → Triage Efficiency"

driver_b: "Support Quality → Communication Style"

relationship: "negative"

description: "Fixing triage (faster processing) may worsen communication (staff rushing more)"

mitigation: "Bundle both interventions to address simultaneously"

bundle_cost_savings: 0.30 # 30% cheaper when bundled

- driver_a: "Clinical Services → Discharge Planning"

driver_b: "Care Coordination → Follow-up Communication"
relationship: "positive"
description: "Improving discharge planning enhances follow-up effectiveness"
bundle_benefit_multiplier: 1.4 # 40% higher combined elasticity

prerequisite_relationships:

- driver: "Digital Services → Patient Portal Usability"
- prerequisite: "Technology Infrastructure → Portal Performance"
- description: "Portal usability improvements require stable infrastructure first"

4.2 Feedback Loops

feedback_loops:

reinforcing:

- name: "Staff Burnout → Service Quality Decline"

drivers:

- "Employee Experience → Workload Management"
- "Clinical Services → Care Quality"

description: "Unaddressed burnout degrades care quality, increasing complaints, increasing workload"

intervention_strategy: "Address root cause (workload) before symptoms (care quality)"

balancing:

- name: "Cost Reduction → Patient Satisfaction Trade-off"

drivers:

- "Operational Efficiency → Resource Utilization"
- "Patient Experience → Service Availability"

description: "Cutting costs may reduce service availability, harming satisfaction"

intervention_strategy: "Model trade-offs before deployment"

Why This Matters

- **Prevents negative side effects:** Agent won't fix triage without addressing communication
 - **Enables bundling:** Agent knows when interventions are cheaper/more effective together
 - **Systems thinking:** Agent identifies root causes vs symptoms
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Module 5: Intervention Library & Historical Performance

Purpose

Define available interventions, expected outcomes, and historical success rates that guide intervention selection.

Required Data

5.1 Intervention Catalog

interventions:

fix_triage_efficiency:

name: "Rapid Triage Protocol Redesign"

applicable_to:

themes: ["Clinical Services", "Emergency Services"]

es_bands: ["ES4_Active_Crisis", "ES5_Critical_Failure"]

opportunity_stream: "Fix"

resource_requirements:

cost: 150000

staff_hours: 640

lead_time_days: 90

deployment_duration_weeks: 12

expected_outcomes:

eri_improvement: 25 # +25 points expected

elasticity_range: [1.8, 2.4]

elasticity_avg: 2.1

success_rate: 0.78 # 78% of deployments succeed

prerequisites:

- "Executive sponsor approval"
- "ED staff engagement sessions completed"

risks:

- "Staff resistance to process changes"
- "Temporary productivity dip during transition"

5.2 Historical Performance

historical_performance:

by_driver_and_intervention:

"Emergency Services → Triage Efficiency":

intervention: "Rapid Triage Protocol"
deployments: 12

outcomes:
successful: 9
partial: 2
failed: 1

avg_elasticity: 2.08
avg_eri_improvement: 23
avg_cost: 142000
avg_duration_days: 87

lessons_learned:
- "Success correlated with ED director engagement"
- "Failures occurred when staff training was rushed"
- "Elasticity highest when bundled with communication improvements"

Why This Matters

- **Evidence-based selection:** Agent picks interventions with proven track records
 - **Realistic expectations:** Agent uses historical elasticity, not theoretical models
 - **Risk mitigation:** Agent knows what went wrong before and how to avoid it
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Module 6: Organizational Context & Governance

Purpose

Define ownership, existing initiatives, approval workflows, and strategic priorities that shape intervention selection.

Required Data

6.1 Ownership Mapping

ownership:
by_theme:
"Clinical Services & Care Delivery":
owner: "Chief Medical Officer"
contact: "cmo@hospital.org"
approval_required: true

approval_sla_days: 14

"Patient Safety & Quality":
owner: "Chief Quality Officer"
contact: "cqo@hospital.org"
approval_required: true
approval_sla_days: 7

"Patient Support & Coordination":
owner: "VP Patient Experience"
contact: "vpx@hospital.org"
approval_required: false

6.2 Existing Initiatives Registry

existing_initiatives:

- initiative_id: "INIT_2025_042"
name: "Shared Decision-Making Program"
owner: "Chief Medical Officer"
affected_drivers:
 - "Consultation Experience → Provider Communication"status: "in_progress"
start_date: "2025-01-15"
end_date: "2025-06-30"
budget: 180000
conflict_risk: "Avoid duplicate efforts on provider communication"

6.3 Approval Workflows

approval_workflows:

tier_1_low_risk:
cost_threshold: 50000
es_bands: ["ES1_Secure_Loyalty", "ES2_Growth_Opportunity"]
approval_required: false
notification_only: true

tier_2_moderate_risk:
cost_threshold: 150000
es_bands: ["ES3_At_Risk"]
approvers: ["VP Patient Experience"]
approval_sla_days: 7

tier_3_high_risk:
cost_threshold: 150000
es_bands: ["ES4_Active_Crisis", "ES5_Critical_Failure"]

approvers: ["CMO", "CQO", "CFO"]
approval_sla_days: 14
board_notification: true

6.4 Strategic Priorities

strategic_priorities:
fiscal_year: 2025

board_mandates:

- priority: "Patient Safety"
weight: 10 # Highest weight
description: "Zero preventable harm events"
- priority: "Emergency Services Excellence"
weight: 8
description: "Top quartile performance on ED wait times"
- priority: "Cost Efficiency"
weight: 6
description: "5% cost reduction without quality impact"

scoring_function: "weighted_sum" # Agent uses weights in optimization

Why This Matters

- **Avoids duplication:** Agent knows CMO is already running a communication initiative
- **Respects governance:** Agent knows what requires approval and from whom
- **Aligns with strategy:** Agent weights interventions by board-level priorities

Module 7: Outcome Metrics & Success Criteria

Purpose

Define what success looks like for each driver, enabling autonomous outcome evaluation and adaptive replanning.

Required Data

7.1 Success Thresholds (By Driver)

success_criteria:

"Emergency Services → Triage Efficiency":

primary_metric: "median_triage_to_assessment_minutes"

baseline: 120

target: 30

minimum_acceptable: 60

secondary_metrics:

- metric: "complaint_volume"

baseline: 47

target: 15

weight: 0.3

- metric: "staff_satisfaction_score"

baseline: 6.2

target: 7.5

weight: 0.2

elasticity_target: 1.5

evaluation_period_days: 90

success_definition: "Primary metric meets target AND elasticity > 1.5 AND change holds for 90 days"

7.2 Evaluation Cadence

evaluation_schedule:

early_signal_check:

timing_days: 30

purpose: "Detect underperformance early"

action_if_failing: "Flag for replanning consideration"

midpoint_evaluation:

timing_days: 45

purpose: "Assess trajectory toward target"

action_if_failing: "Pause intervention, reallocate resources"

final_evaluation:

timing_days: 90

purpose: "Measure elasticity and durability"

action_if_successful: "Store in IML, scale if appropriate"

action_if_failed: "Root cause analysis, store learnings"

Why This Matters

- **Autonomous evaluation:** Agent knows what to measure and when
 - **Adaptive replanning:** Agent can pause underperforming interventions mid-cycle
 - **Learning:** Agent knows what "success" means and stores appropriately
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Integration: How the Agent Uses This Knowledge Base

Portfolio Optimization Algorithm (Pseudocode)

```
def optimize_portfolio(  
    seus: List[SEU],  
    constraints: ResourceConstraints,  
    risk_models: RiskModels,  
    dependencies: DependencyGraph,  
    interventions: InterventionLibrary,  
    org_context: OrganizationalContext  
) -> List[OU]:  
  
    # Step 1: Filter SEUs by priority and risk  
    candidates = []  
    for seu in seus:  
        # Calculate risk-adjusted score  
        base_score = seu.priority_score  
  
        # Apply risk multipliers  
        if seu.has_regulatory_risk(risk_models):  
            base_score *= risk_models.regulatory_multiplier  
  
        if seu.has_reputational_risk(risk_models):  
            base_score *= risk_models.reputational_multiplier  
  
        if seu.has_churn_risk(risk_models):  
            churn_value = seu.churn_probability * seu.revenue_at_risk  
            base_score += churn_value / 1000 # Normalize to score scale  
  
        # Apply strategic priority weights  
        theme_weight = org_context.strategic_priorities.get(seu.theme)  
        base_score *= theme_weight  
  
    candidates.append({  
        'seu': seu,  
        'risk_adjusted_score': base_score
```



```
)
```

```
# Step 2: Rank candidates
candidates = sorted(candidates, key=lambda x: x['risk_adjusted_score'],
reverse=True)

# Step 3: Select interventions within constraints
selected = []
budget_used = 0
staff_hours_used = 0

for candidate in candidates:
    seu = candidate['seu']

    # Get applicable interventions
    applicable = interventions.get_for_seu(seu)

    # Select intervention with highest expected elasticity
    best_intervention = max(applicable, key=lambda i: i.expected_elasticity)

    # Check budget constraint
    if budget_used + best_intervention.cost > constraints.budget:
        continue

    # Check staff capacity constraint
    if staff_hours_used + best_intervention.staff_hours > constraints.staff_hours:
        continue

    # Check for dependencies
    bundled = dependencies.check_bundles(seu, selected)
    if bundled:
        # Adjust cost for bundle savings
        actual_cost = best_intervention.cost * (1 - bundled.cost_savings)
        selected.append({
            'seu': seu,
            'intervention': best_intervention,
            'cost': actual_cost,
            'bundled_with': bundled.drivers
        })
    else:
        selected.append({
            'seu': seu,
            'intervention': best_intervention,
            'cost': best_intervention.cost,
            'bundled_with': None
```

```
})
```

```
budget_used += best_intervention.cost  
staff_hours_used += best_intervention.staff_hours
```

```
return selected
```

Deployment Guide

1. Initial Setup

Create knowledge base directory structure

```
emotional_infrastructure/  
knowledge_base/  
  financial_constraints.yaml  
  capacity_constraints.yaml  
  risk_models.yaml  
  dependencies.yaml  
  intervention_library.yaml  
  org_context.yaml  
  success_criteria.yaml
```

```
metadata.yaml # Versioning, last_updated, owner
```

2. Population Strategy

- **Month 1:** Populate financial constraints (budget data from finance)
- **Month 2:** Populate capacity constraints (staff data from HR)
- **Month 3:** Populate risk models (regulatory requirements, historical churn)
- **Month 4:** Populate dependencies (workshop with operational leaders)
- **Month 5:** Populate intervention library (historical project data)
- **Month 6:** Populate org context (ownership mapping, approval workflows)
- **Month 7:** Populate success criteria (KPI definitions from quality team)

3. Maintenance Cadence

- **Quarterly:** Update budget allocations, staff capacity, strategic priorities
- **Annually:** Refresh intervention library with new historical performance data
- **As-needed:** Update dependencies when new drivers or initiatives emerge

4. Governance

- **Owner:** Chief Experience Officer or VP Patient Experience
 - **Approvers:** CFO (financial), CHRO (capacity), CQO (risk), CMO (clinical dependencies)
 - **Version control:** Git-based versioning with approval workflows
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Benefits: What This Enables

For the Agent

1. **Autonomous portfolio optimization** within real-world constraints
2. **Risk-weighted decision making** that balances ROI with regulatory/reputational exposure
3. **Dependency-aware bundling** that prevents negative side effects
4. **Multi-period planning** that sequences interventions across quarters
5. **Adaptive replanning** that responds to changing conditions mid-cycle

For the Organization

1. **Maximized ROI:** Agent selects interventions that maximize elasticity per dollar
2. **Risk mitigation:** Agent prioritizes regulatory/safety risks appropriately
3. **Resource efficiency:** Agent respects capacity limits, avoiding overload
4. **Strategic alignment:** Agent weights decisions by board-level priorities
5. **Transparency:** Every decision traces back to constraints, risks, and evidence

For Stakeholders

1. **Predictable planning:** Leaders know what's affordable and achievable each quarter
 2. **Justified decisions:** Every intervention selection has documented reasoning
 3. **Reduced conflict:** Bundling and ownership mapping prevent duplicate efforts
 4. **Continuous learning:** Historical performance data improves future decisions
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Conclusion: From Substrate to Strategy

The **SEU** provides the perception and decision substrate—making emotion visible, comparable, and prioritizable.

The **Resource Constraints Knowledge Base** provides the strategic layer—making decisions feasible, risk-aware, and optimized within real-world limits.

Together, they enable **fully autonomous agentic action**: an AI system that can perceive emotional truth, decide what matters most given constraints, act with disciplined execution, and remember what worked—without human intervention at any step.

This is the missing piece that transforms emotional infrastructure from decision support into autonomous intelligence.

Next Steps

1. **Start simple**: Populate financial constraints first (budget, cost models)
2. **Iterate**: Add capacity constraints once financial optimization is working
3. **Layer complexity**: Add risk models, dependencies, intervention library progressively
4. **Validate**: Compare agent decisions to human decisions; refine weights and thresholds
5. **Scale**: Once validated in one domain (e.g., healthcare), generalize to others (commerce, education, government)

The architecture is ready. The knowledge base makes it operational.